Customer Segmentation With Clustering

Stakeholder Report

Prepared By: Israel Olujobi

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University of Cambridge

FourthRev

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1. Problem Overview

An e-commerce company is looking to understand its customers and improve its marketing efforts by performing customer segmentation with clustering models. Its data set is transnational and includes customers from 47 countries, across five continents (Oceania, North America, Europe, Africa, and Asia).

The dataset contains information unique to the customer including Quantity, Customer ID, Order ID, Unit Cost, etc. To aid a more comprehensive and channeled marketing approach, features critical for effective customer segmentation and modelling are derived. These include **Frequency**, **Recency, Customer lifetime value (CLV), Average unit cost, Customer age.** Poor data quality and management, choosing the wrong or irrelevant criteria, ethical and legal implications relating to collection and analysis of data, finding the right balance when clustering segments, are some of the issues which could negatively impact the understanding of the business scenario for effective customer segmentation.

2. Overview of Approach

The methodology used was majorly centered around unsupervised machine learning models (K-means clustering, Hierarchical clustering) and unsupervised evaluation models (Elbow method, Silhouette score) (**Table 1**).

Model	Type	Parameters included	Key Information
K-means	Unsupervised ML	Number of Clusters (k)	Assigns data points to clusters based on the shortest distance between each data point and each centroid
Elbow Method	Unsupervised evaluation model	n_clusters, WCSS	The bend in an elbow plot represents the optimal number of clusters
Silhouette score	Unsupervised evaluation model	n_clusters, silhouette score	Score ranges from -1 to 1
Hierarchical clustering	Unsupervised ML	Linkage, method	Visualises clusters in a dendrogram.

Table 1: Unsupervised ML and evaluation methods applied.

2.1. Customer Segmentation Workflow

The end-to-end process used for clustering the transnational customer dataset is visualized below (**Figure 1**). This included preprocessing the raw data, using feature engineering to derive the important features, analyzing the descriptive statistics, and visualizing the data via boxplots and histograms. Detecting the outliers and feature scaling followed.

The number of clusters to be used for clustering the dataset was determined using the elbow method, silhouette score and dendrogram. All were assessed individually and results were combined to determine the optimum number of clusters.

K-means clustering was undertaken using the number of clusters chosen and results were used to segment each customer in the dataset into clusters, additionally visualized by boxplots.

PCA and t-SNE provided data visualization for both machine learning models through dimensionality reduction.

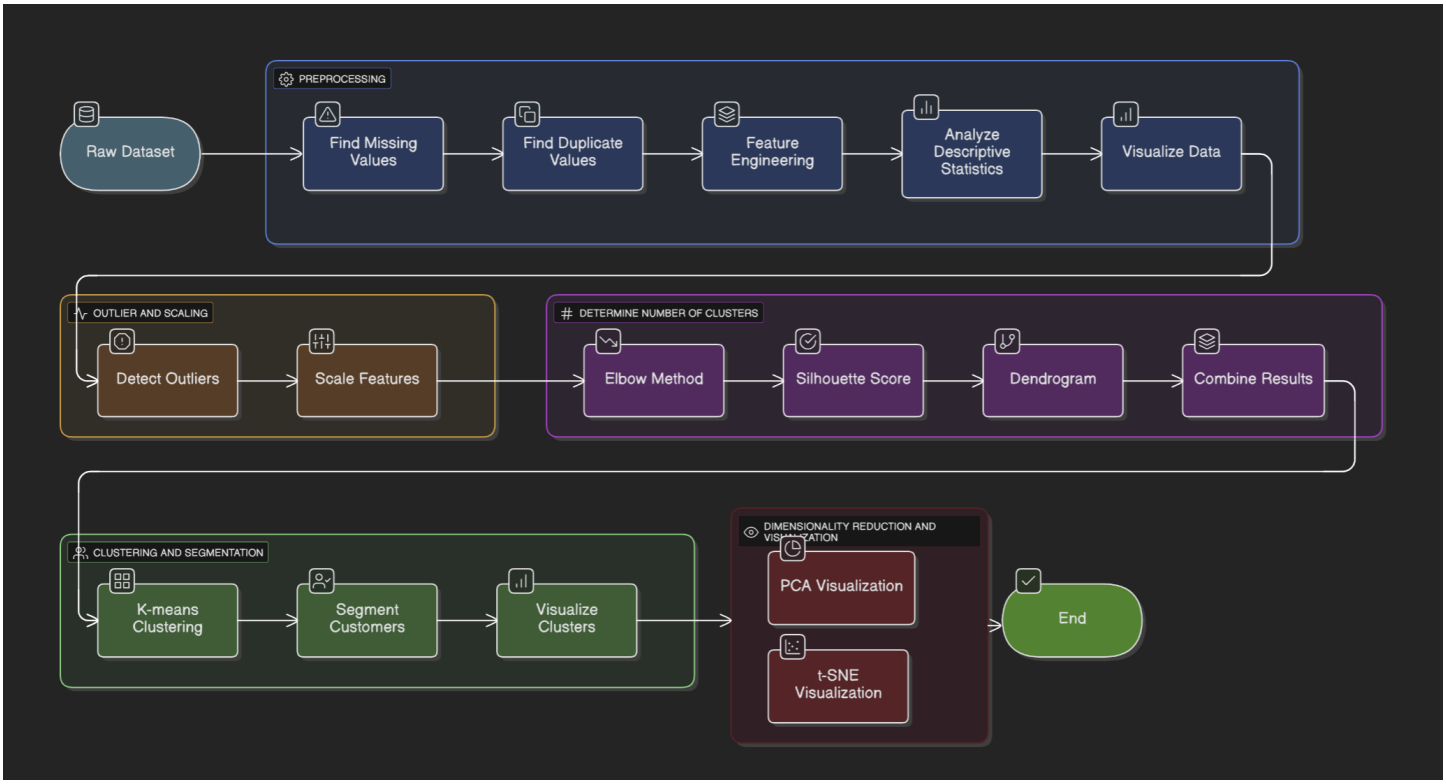

Figure 1: A high-level visualization of the customer segmentation pipeline.

2.2. Data and Features

The initial transnational dataset included 951,669 data points in the data set, each with 20 features. Using feature engineering, these features were narrowed into 5 main features namely: *frequency, recency, avg_unit_cost, CLV and age.*

The box plot visualization of the data reflected clear outliers largely focused above the top whisker of each feature. Additionally on the right-hand side of the plot, right-skewness is observed in the distribution of majority of the features. (**Figure 2**).

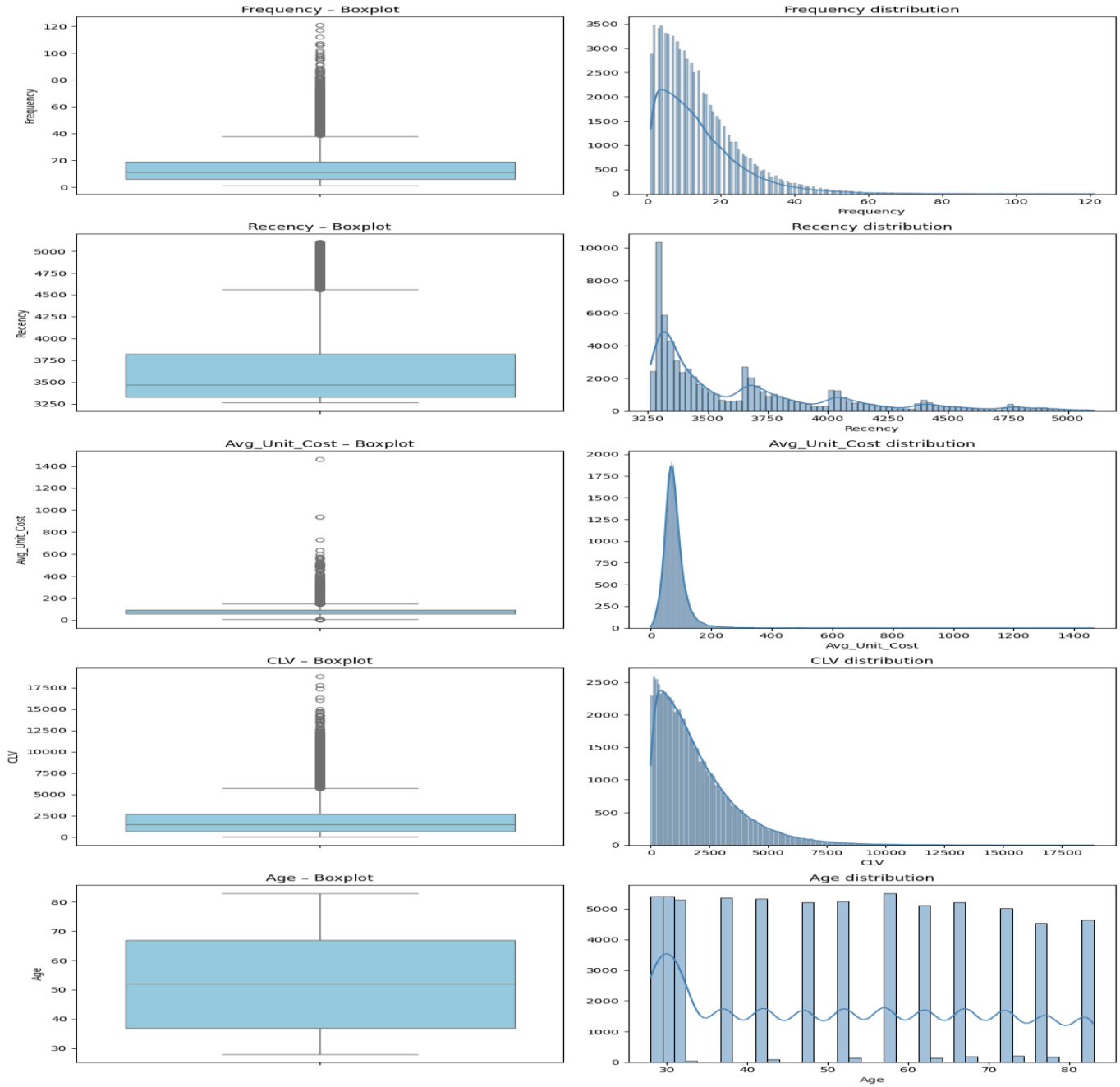

Figure 2: Visualization of feature engineered data.

2.3. Outliers

It is important to note the outliers observed using IQR. The data table below **(Table 2)** confirms the observations from the boxplots. All features except Age contain statistical outliers.

Recency contains the highest number of outliers, at ***3353 outliers (4.91%)***.

Feature	Outliers	% of feature
Recency	***3,353***	***4.91%***
Average Unit Cost	*2,889*	*4.23%*
Frequency	*2,654*	*3.89%*
CLV	*2,590*	*3.79%*
Customer Age	*0*	*0.00%*

Table 2: Observed outliers per feature.

3. Interpretation of Customer Segmentation Results

Elbow Method: Based on the Elbow plot **(Figure 3)**, there is ***no clear bend/elbow*** in the diagram, making it difficult to outrightly determine an optimal cluster, thus other evaluation methods needed to be applied. Despite this between 4-5 clusters can be estimated as the optimum clusters from the plot.

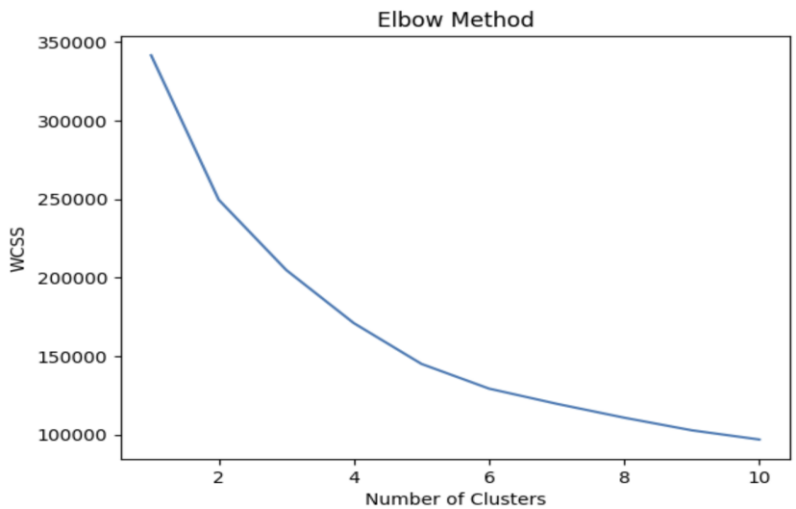

Figure 3: Elbow method plot.

Silhouette score method: Based on the Silhouette scores and plot generated below, n_clusters = 5 appears to be the ideal number of clusters, as it holds the ***highest silhouette score (0.26711)*** **(Table 3)**, there is a stable distribution across all clusters, despite the presence of outliers. It also has more compact clusters with less variability amongst its points **(Figure 4)** compared to n_clusters = 2, which has a closer silhouette score.

Further observations also reflect silhouette scores falling below zero in all clusters, indicating the presence of outliers in the data set, as indicated in earlier analysis methods.

n_clusters	Silhouette score
5	***0.26711***
2	*0.2607*
4	*0.2532*

Table 3: Silhouette scores per n_cluster.

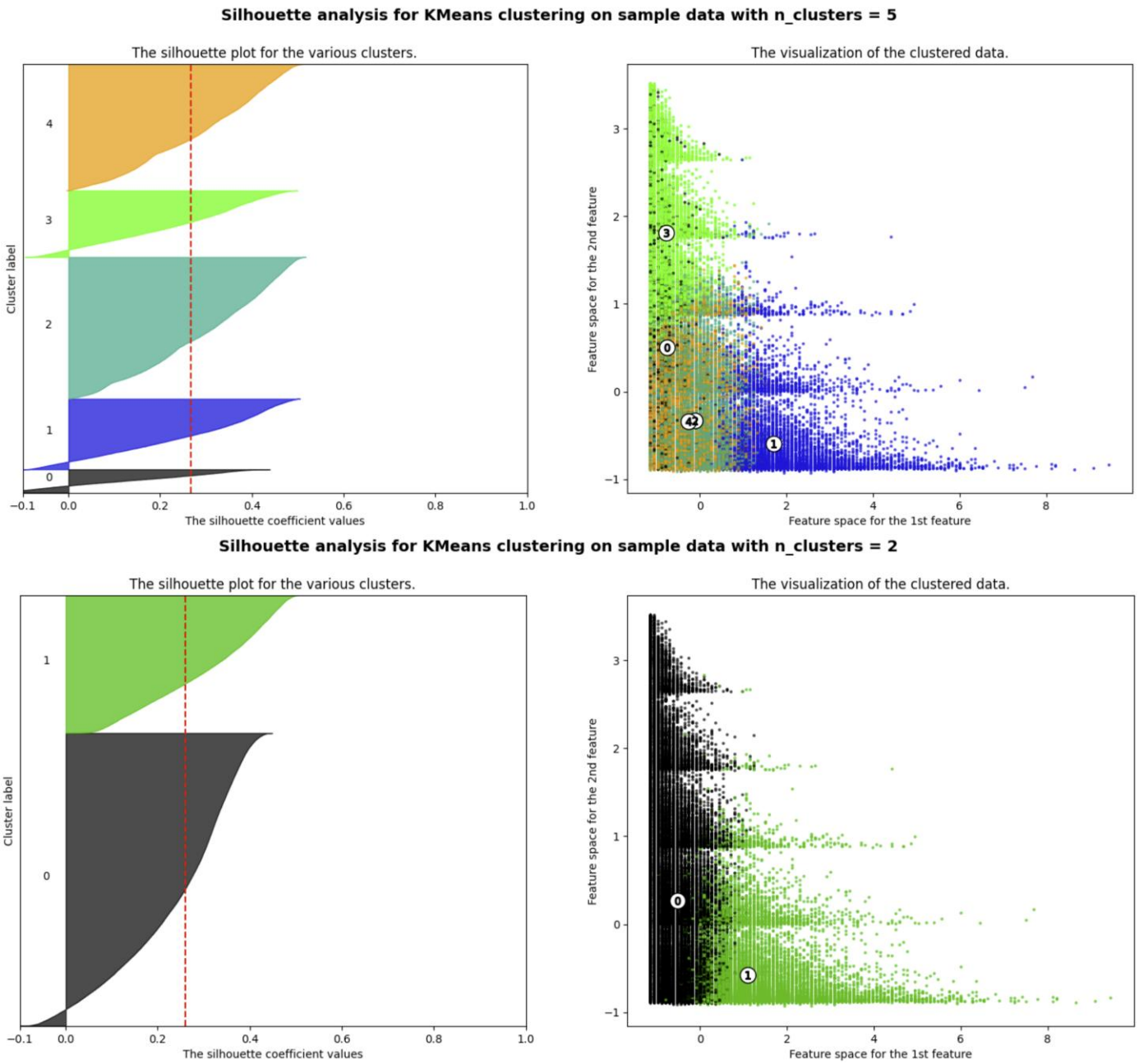

Figure 4: Visualization of Silhouette analysis.

Hierarchical Clustering: Based on the dendrogram below **(Figure 5)**, a ***cutoff at 150*** represents the ideal number of clusters at 5. The distance tolerance in selecting 5 clusters is ideal, as reducing/increasing the cutoff slightly doesn't increase/reduce the number of clusters significantly.

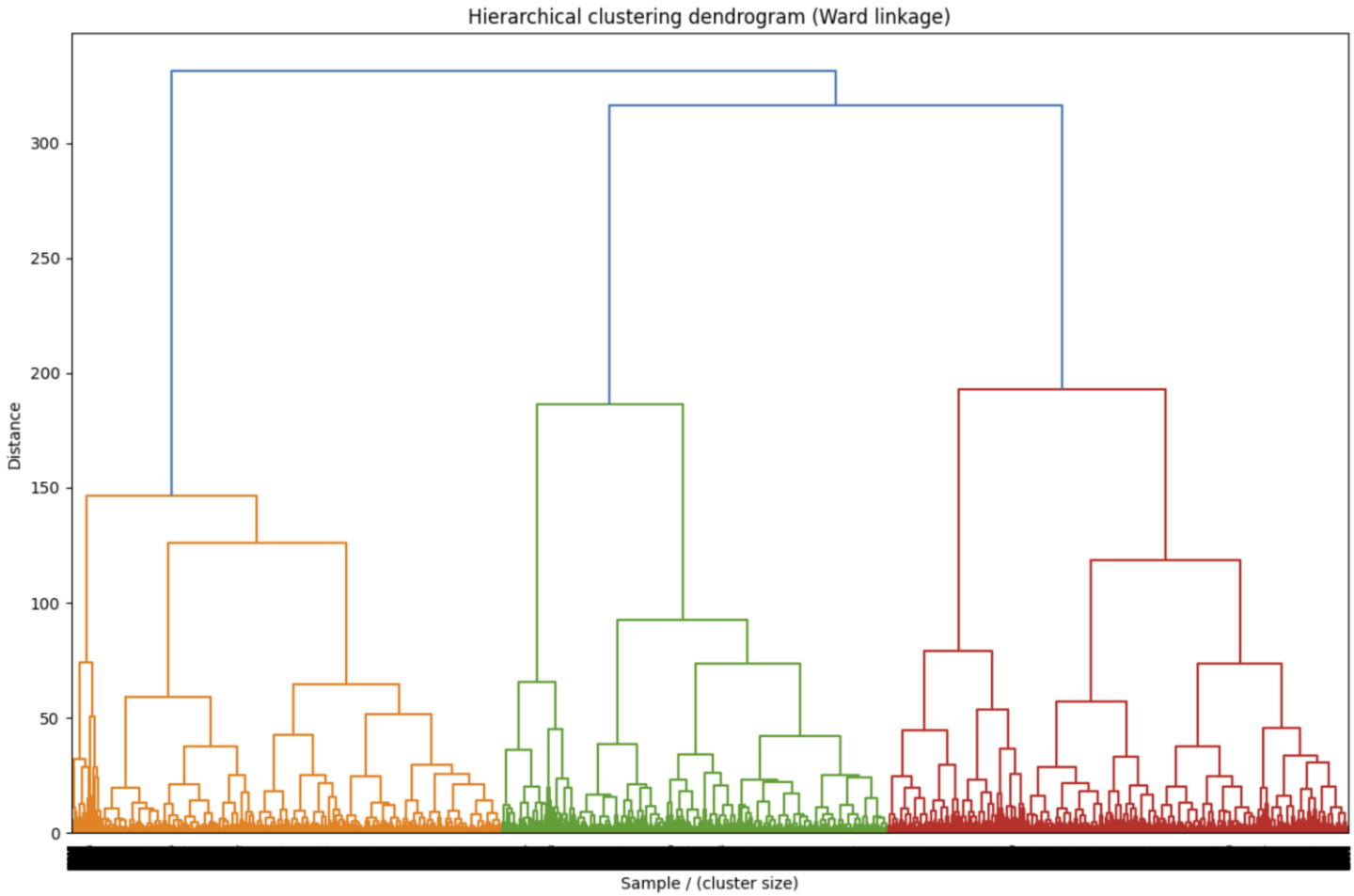

Figure 5: Visualization of Hierarchical clustering dendrogram.

K-means Clustering: The combined observations from the 3 methods above reflect ***n_clusters = 5*** as the optimum number of clusters, this was used for k-means clustering. The observed k-means plot was plotted, on two features, frequency and recency.

Based on the plot observations **(Figure 6)**, there are 5 distinct and clearly separated groups that align more closely with expected patterns, noting that data points appear closely aligned around assigned centroids.

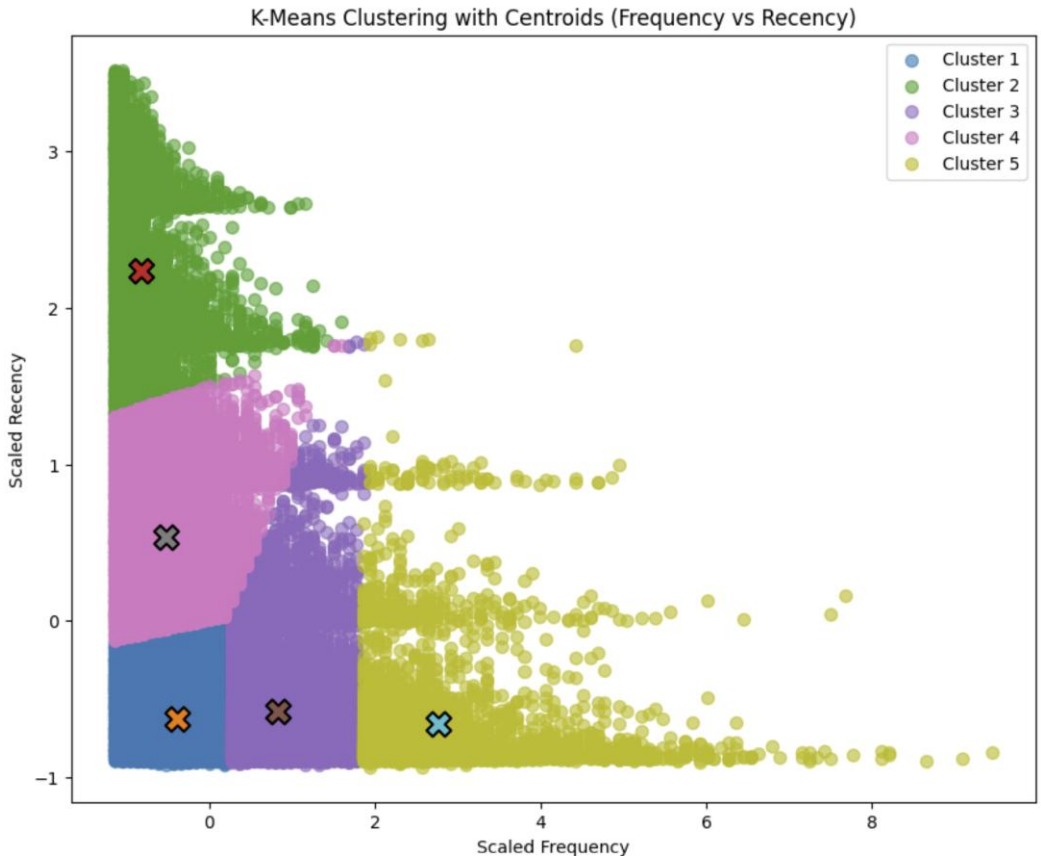

Figure 6: Visualization of K-means clustering.

4. Visualization of Customer Segmentation

The clusters which were trained on frequency and recency features, were used to place each customer in segments ranging from 0-4 and visualized with regards to each feature.

The boxplot below reflects the clusters **(Figure 7).** A notable observation lies in ***cluster 4***, which has a higher median in frequency and CLV, a low median in recency and avg_unit_cost. These are the active customers, as they buy often and recently. In this case the company should focus on marketing techniques to contribute to increase in engagement, raise the median and reduce outliers. Providing reward or referral programs could boost this initiative etc.

In ***cluster 1***, we observe that median in recency is significantly higher, whilst the medians in the rest of the features are low. In this case the company should focus on promoting targeted offers to prevent customers from drifting further away.

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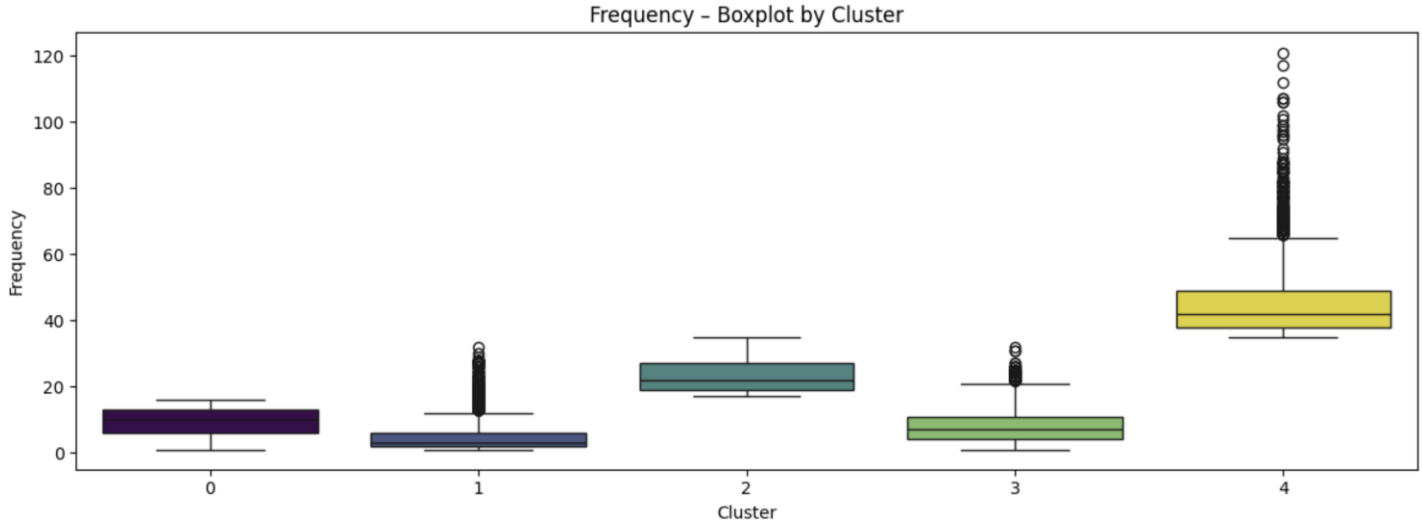
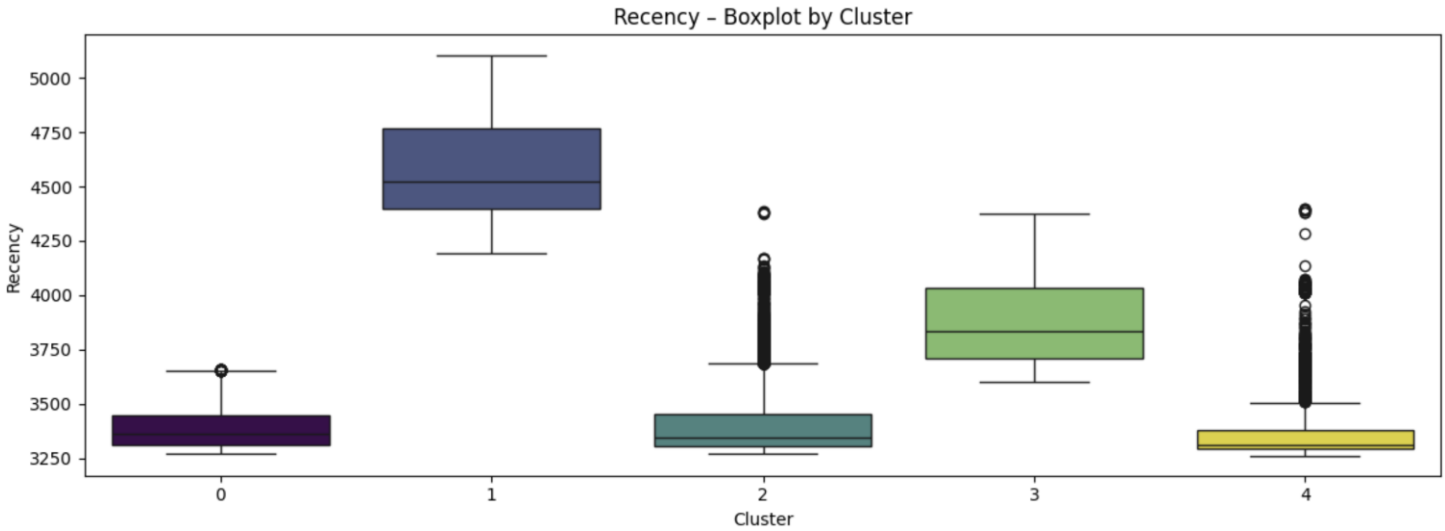
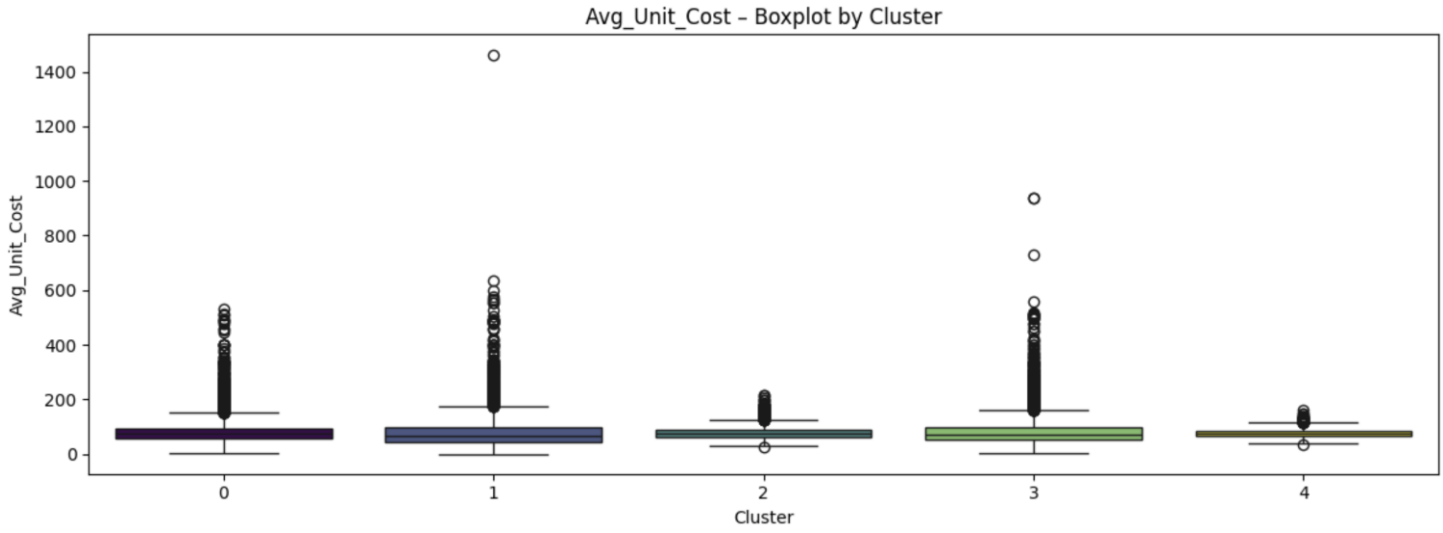

Frequency – Boxplot by Cluster
Frequency
Cluster
0
1
2
3
4
0
20
40
60
80
100
120


Recency – Boxplot by Cluster
Recency
Cluster
0
1
2
3
4
3250
3500
3750
4000
4250
4500
4750
5000


Avg_Unit_Cost – Boxplot by Cluster
Avg_Unit_Cost
Cluster
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

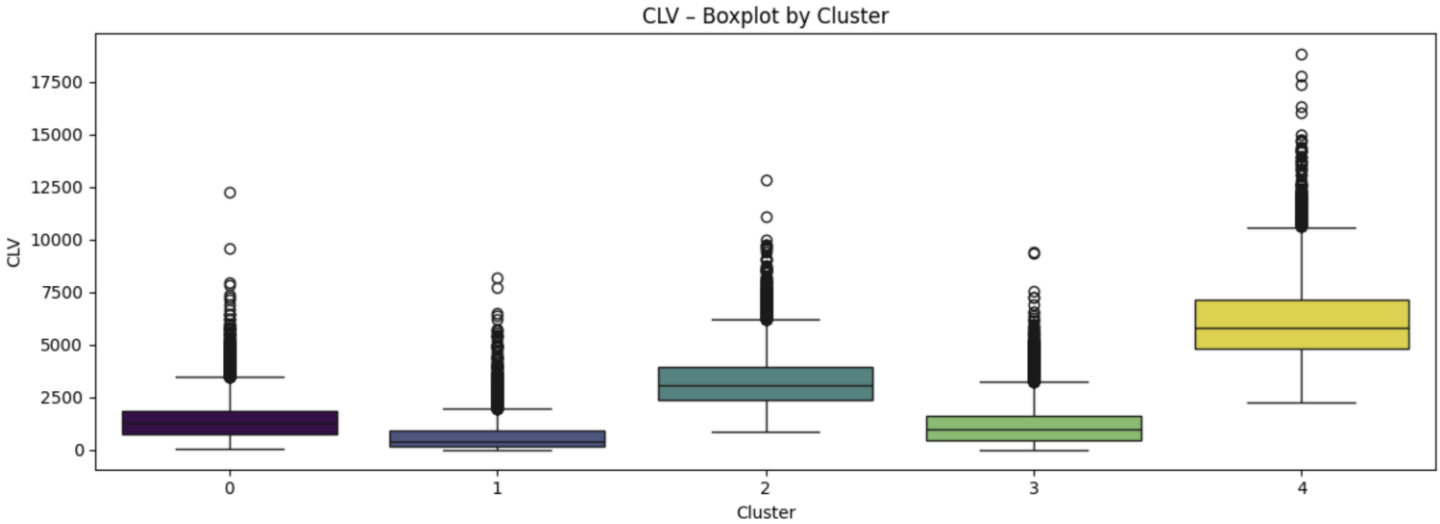


Figure 7: Visualization of Customer Segmentation.

5. Evaluation of 2D PCA and t-SNE Plots

PCA: was used to visualize the 5 cluster points in the segmented data whilst projecting the data onto a two-dimensional plane (**Figure 8**). Despite some overlaps in the clustering, we observe the significant separation among the clusters confirming our earlier analysis.

Note that there is an extreme case at the top left of the chart. This could be an extreme/one-off case and requires further investigation and analysis.

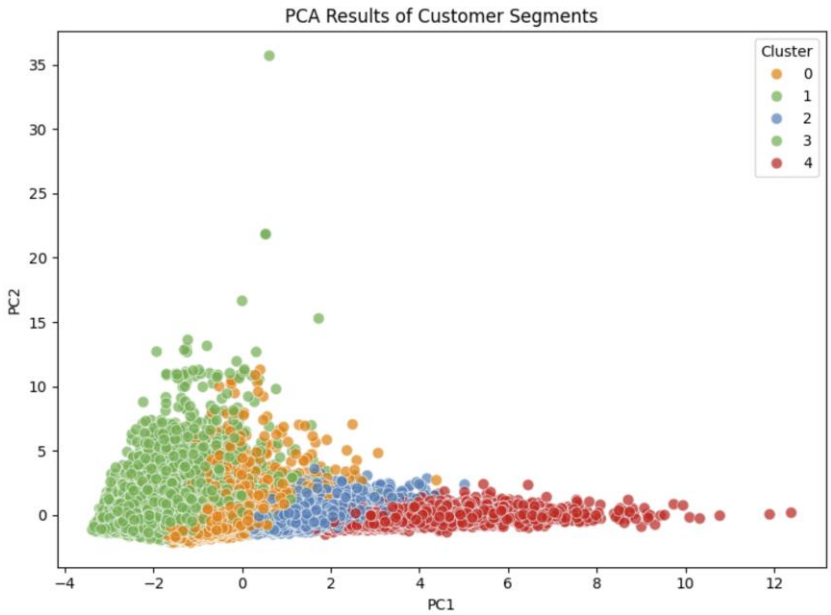


Figure 8: 2D PCA visualization.

t-SNE: is a non-linear dimensionality reduction technique, much more suited for data visualization than PCA. We observed different perplexities to view different lenses of the clustering plot.

From the plot below (Figure 9), ***perplexity = 15*** reflects the tightest, most compact and most differentiated clusters out of the 3 perplexities.

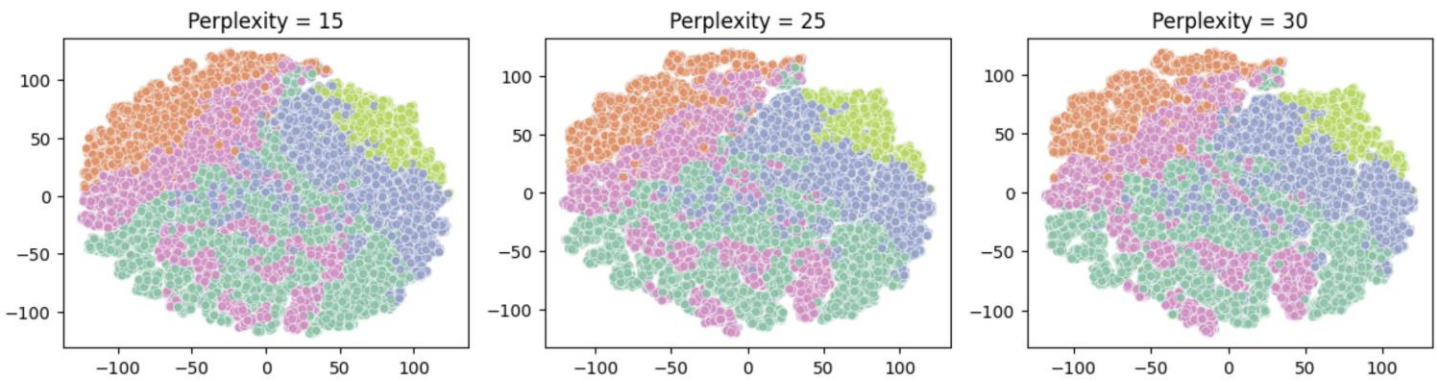


Figure 9: 2D t-SNE visualization.

6. Conclusions and Recommendations

- Based on the analysis carried out, the 5 features derived from feature engineering was the bedrock of the entire model. 5 clusters were determined as the optimal number of clusters which was used to effectively segment customers into clusters and draw observations on the visualized data.

- Observations reflect that the e-commerce company majorly deals in low-cost goods and with customers within average age (45-55). The PCA and t-SNE plots further confirmed the robustness and clear differentiation of each cluster.

- Clustering results of the e-commerce's company transnational data set aided clearer visualization and provided key insights into the company's customer base and future marketing focus.

7. Next Steps

- The e-commerce company should use this robust customer segmentation model to assist in generating key marketing strategies that would efficiently target the needed customer base and effectively serve its customers better. This will help the company to have a more customer-centric focus, improving marketing efficiency.

- Findings should be duly conveyed to business stakeholders.

8. References

- SAS, 2024. CUSTOMERS_CLEAN [Data set]. SAS. Last revised on 15 December 2021. [Accessed 20 February 2024].